

Chapter -1 (Charge and Electric potential)

1. Define the following:
 - a. Electric Field & Electric charge
 - b. Electric Potential
 - c. Electric Dipole and dipole moment
 - d. Electric flux
2. State and explain Coulomb's law for electrostatics
3. Derive an expression for an electric field due to a long uniformly charged wire.
4. Find the electric field at a point on the axis of a charged circular turn wire.
5. Find an expression for the electric potential and electric field due to an electric dipole.
6. Derive an expression for electric potential at a distance r from point charge q .
7. State and explain Gauss's law for electrostatics
8. Related problems

Chapter -2 (Magnetic Field)

1. Define Magnetic Induction
2. Find the magnetic induction at any point on the axis of solenoid
3. Find the magnetic field due to the current flowing in a straight wire
4. Explain self-induction/ inductance. Deduce a mathematical expression for the self-inductance of Solenoid.
5. State Biot Savart law. Find the value of B due to a current loop of radius a carrying current I at a distance x on the axis from the centre of the loop. What will be the value of B at the centre of the loop
6. State the Faraday's law of electromagnetic induction.
7. Define coefficient of self induction of a coil.
8. State the laws of Faraday and Lenz for electromagnetic induction
9. Define mutual inductance between a pair of coils. Deduce a mathematical expression for the mutual inductance between two coaxial solenoid.

Chapter-3(Current and Resistance)

1. What is RC circuit. Obtain an expression for the growth charge and current when a capacitor is charged through a resistance for a constant emf.
2. A capacitor is charged for a constant emf and then allowed to discharge through a resistance. Find the expression for the discharge of the capacitor.
3. State and explain Ohm's Law
3. Define the following
 - a. Electric current
 - b. Current density
 - c. Capacitor & capacitance
 - d. Resistor, resistivity
 - e. Conductor, conductance and conductivity